Siyuan Chang

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Education

Renmin University of China, School of Statistics Sep 2024 – Present
Data Science and Big Data Technology
• GPA: 3.9/4.0, Top 5% in major
• **Programming:** Proficient in C++ and Python; full score in Data Structures
• **Mathematics:** Full scores in Mathematical Analysis I, II & III, Advanced Algebra, and Probability Theory
• **Language:** CET-4: 688, CET-6: 656; full scores in all college English courses.second foreign language: French
• **Prize:** National Third Prize, National Mathematics Competition for College Students, 2025
• **Honor:** First-class Academic Excellence Scholarship, RUC, 2025

Academic Papers

• WED-Net: A Weather-Effect Disentanglement Network with Causal Augmentation for Urban Flow Prediction.
Qian Hong, **Siyuan Chang**, Xiao Zhou.
arXiv:2601.22586. *WWW 2026*.
• Controllable Affective Generation via Latent Vector Steering.
Xixian Yong, **Siyuan Chang**, Xiao Zhou, Zihe Wang, Yingying Zhang,Yefeng Zheng, Xian Wu.
EMNLP 2026.

Research Experiences

• Advisor: Xiao Zhou, Gaoling School of Artificial Intelligence, Renmin University of China
• **WED-Net: A Weather-Effect Disentanglement Network with Causal Augmentation for Urban Flow Prediction**
· Disentangle weather-induced traffic patterns via attention mechanisms, with a weather discriminator and a causal data augmentation strategy that preserves causal structures while perturbing non-causal components.
• **Controllable Affective Generation via Latent Vector Steering**
· Propose EmoVec, a post-hoc framework for controllable affective generation in llms via latent vector steering, addressing the emotional flattening issue caused by alignment techniques without weight tuning or retraining

Research Interests

• **Machine Learning**
· Pursuing generalizable knowledge discovery from **unlabeled or sparsely annotated data**.
· Embedding **graph-structured** data into low-dimensional vector spaces while preserving topological properties.
• **Data Mining**
· Extracting predictive and descriptive patterns from **sequential time-stamped** data.
· Integrating **heterogeneous information** across text, image, graph for robust knowledge extraction.
• **Optimization**
· Analyze **stochastic optimization** for robust decision-making in uncertain supply chain systems.
· Investigate structural optimization to enhance the theoretical efficiency of **recommendation models**.
• **Social Computing**
· Building data-driven and AI-enabled frameworks to optimize **urban traffic** operation and management.
· Revealing the **diffusion laws of information** and viewpoints in online social systems.
· AI-powered **sentiment analysis** from textual and multimodal data.